MGMT 59000 - Machine Learning
02/23/2026

Learning When to trust Signals

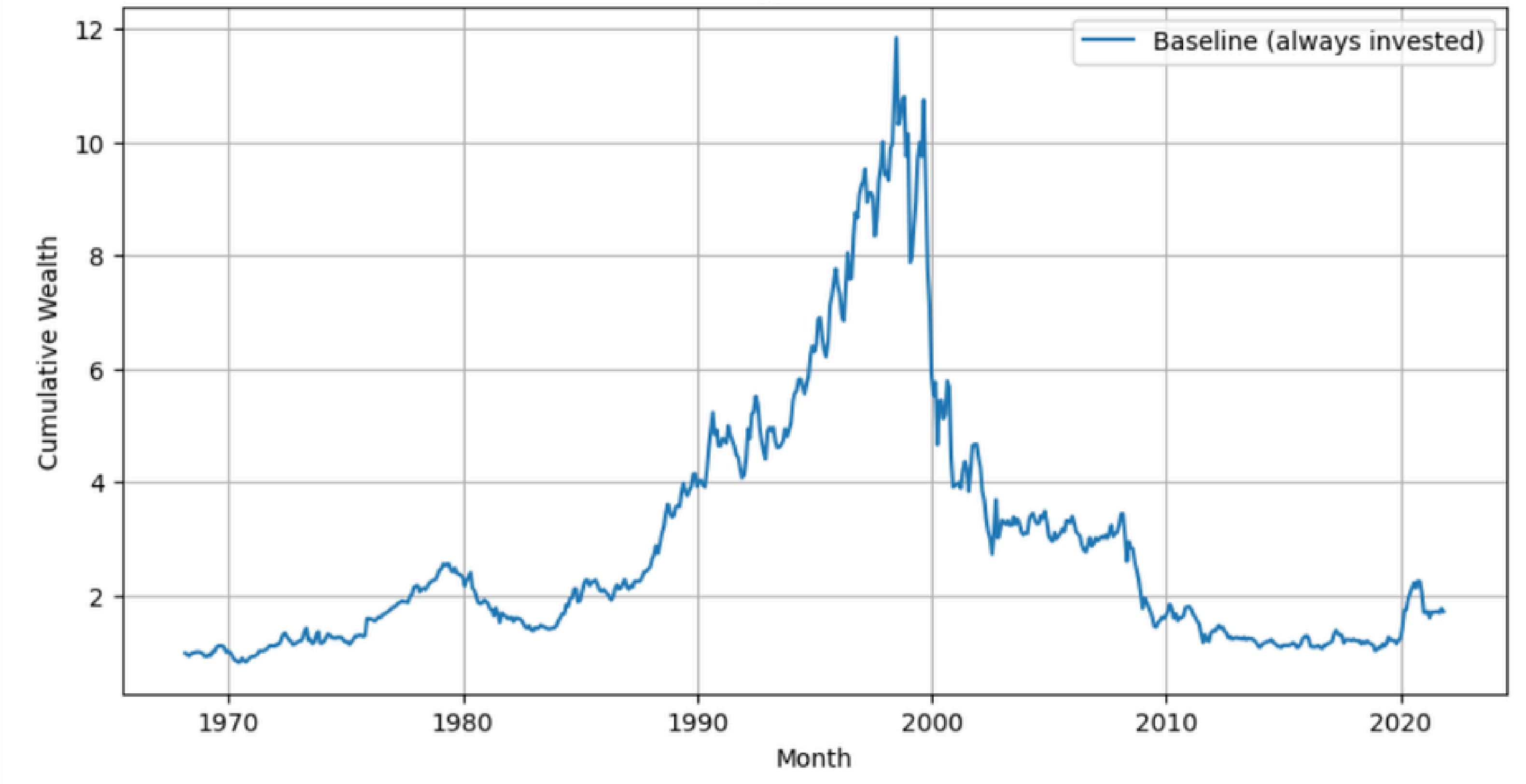
A Macro-Driven Gating Framework for Regime-Aware Trading

MGMT 59000 - Machine Learning
Syed Bilal Farrukh
Mishaal Kamran Hussain

Motivation

Financial return prediction is a weak-signal problem where performance depends on market conditions, with signals often breaking down during volatile periods despite working well in stable regimes.

Drawdowns destroy wealth



Research Question

"Can a simple gating model improve performance by learning when to act on predictions?"

Sub-Design Questions

1. **Fixed vs rolling training**: adaptivity vs stability
2. **Binary vs soft gating**: discrete vs continuous exposure
3. **Feature choice**: macro (stable) vs diagnostics (noisy)
4. Why does regime separation fail, yet gating still work?

Conceptual Framework

Baseline Model

What to trade
(signal)

Decision (Gating) Layer

Whether to trade
(confidence)

Portfolio Action

Trading Decision

- Learns cross-sectional return signals
- Ranks assets (long-short portfolio)
- Optimized for statistical performance

- Estimates $P(R_{t+1} > 0)$ using macro & signal diagnostics
- Controls exposure (trade vs no trade)

Why this matters

Small, noisy prediction signals can still improve outcomes when translated into **selective decision rules**

Baseline Model

A long-short strategy that always acts on model predictions, without learning when those predictions should be trusted.

Basic Model Design | Multi-Tower Neural Network

```
Macro → Tower \
Firm  → Tower  → Fusion → Return Prediction
SIC   → Tower /
```

Design Choice	Why?
Firm tower (128–64), macro & industry (32)	Captures dominant firm-level signal; limits macro overfitting
Fusion head (64–32)	Learns cross-feature interactions without excess complexity
Training = 5 epochs	Reduces overfitting in noisy financial data
Weighted MSE loss	Aligns with value-weighted portfolio objective

Market-cap weighted loss function

$$\mathcal{L} = \sum_i w_{i,t}(\hat{r}_{i,t+1} - r_{i,t+1})^2$$

where $w_{i,t} \propto \mathrm{MarketCap}_{i,t}$

Prioritizes errors on economically significant assets

Time-Series Rolling Validation | To avoid look-ahead bias

Train on the past 10 years of data and predict the next year, rolling the window forward each year to generate strictly out-of-sample predictions

Portfolio Construction | Long top decile, Short bottom decile

Key Assumption

The strategy remains fully invested at all times, implicitly assuming that model signals are equally reliable across different market conditions.

Meta-Model

Logistic Regression Classifier with a lightweight regime-gating layer

Basic Model Design

```
Macro State + Signal Diagnostics
            ↓
     Logistic Classifier
            ↓
       Exposure Gate
```

Inputs: macro state + signal diagnostics

Trained via binary cross-entropy (time-ordered split)

Gating Rule: Trade only if probability ≥ threshold, otherwise reduce / remove exposure

Research Focus

Whether signals are reliable across regimes

For this, we would change:

- Fixed vs Rolling Training
- Exposure function (binary vs soft gating)
- Threshold sensitivity
- Regime detection inputs

What we Learn

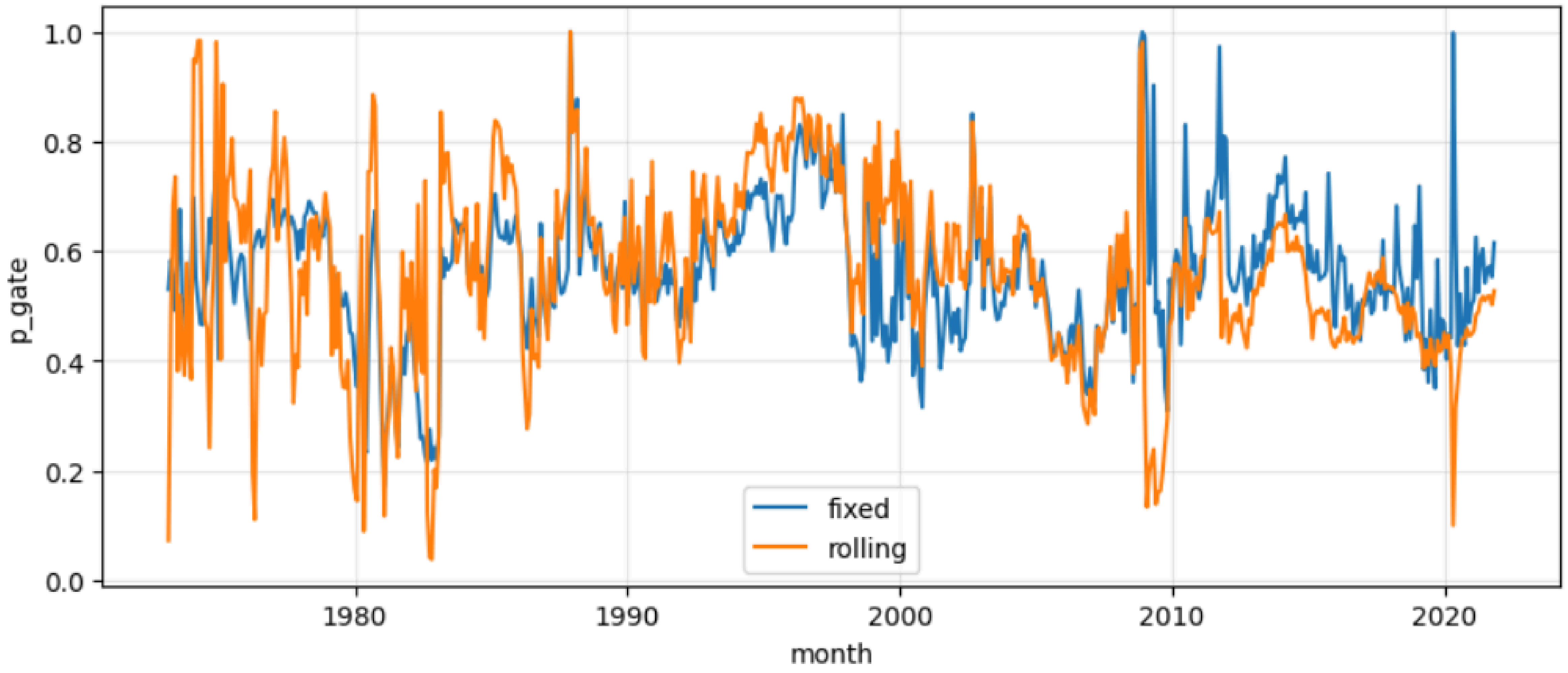
- How strict exposure control affects risk vs return
- Whether soft or hard gating better stabilizes performance
- How signal reliability varies across market regimes

DESIGN ELEMENT 1 FINDINGS

Fixed vs Rolling Training

Why Adaptivity Can Hurt Performance in Noisy Environments

Gate Probability over time | Fixed vs Rolling Training



Insights

- Both models produce noisy probabilities!
- Rolling shows larger, sharper swings over time
- Fixed changes more gradually and consistently

Tradeoffs

Rolling More adaptive, but amplifies short-term noise

Fixed More stable, but less responsive to regime changes

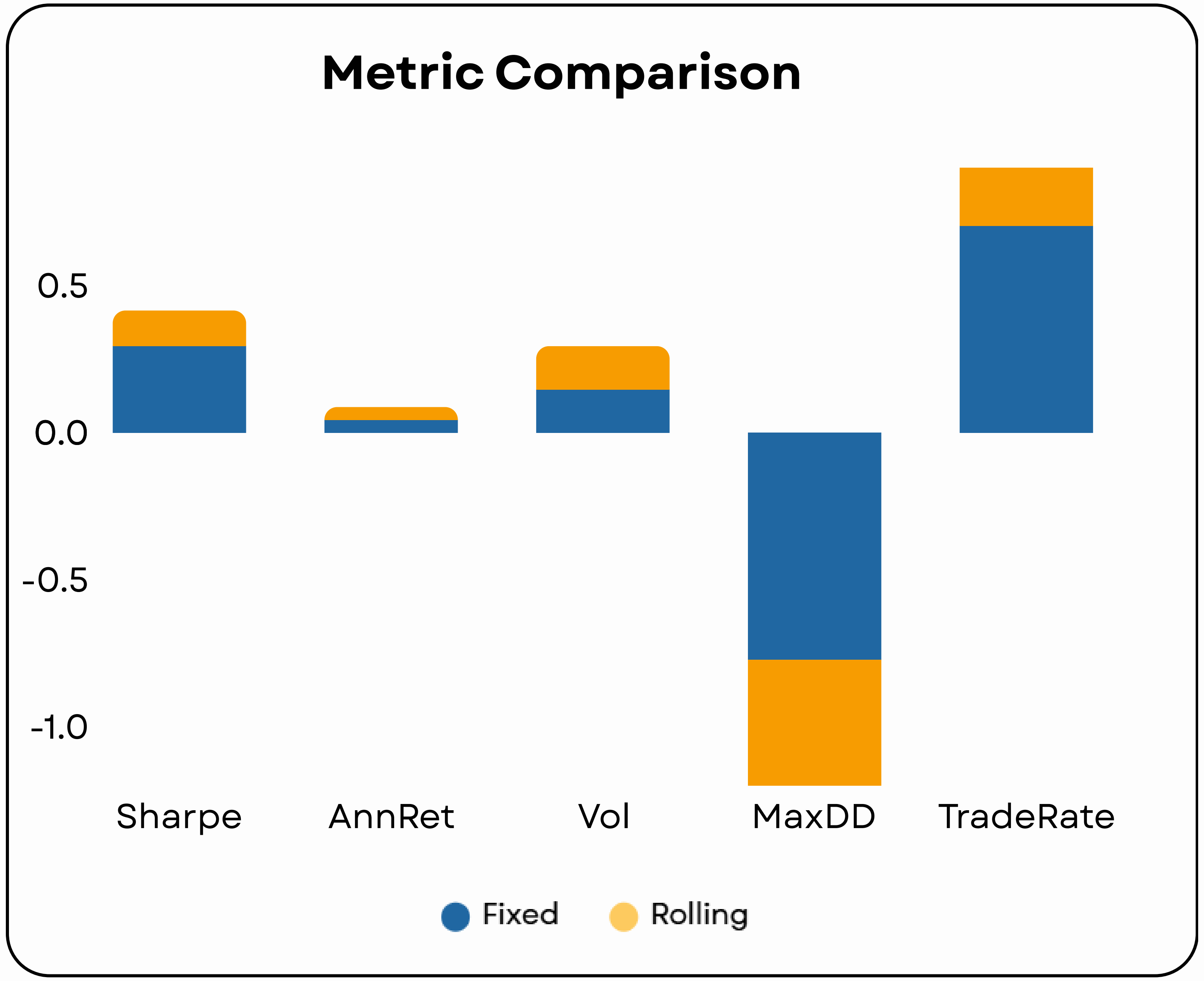
Limitations

Fixed may lag in new market conditions whereas Rolling is highly sensitive to noise and limited data

DESIGN ELEMENT 1 FINDINGS

Fixed vs Rolling Training

We fix the feature set (both), use binary gating with τ = 0.50, and keep portfolio construction unchanged, varying only the training scheme



Metric Comparison

Model	Sharpe	AnnRet	Vol	MaxDD	TradeRate
Fixed	0.295	0.0434	0.1469	-0.7727	0.7025
Rolling	0.12	0.0434	0.1469	-0.8294	0.595

Insights

Sharpe Ratio — Rolling leads to more volatile and less stable Sharpe, reducing risk-adjusted performance

Annualized Return — Similar returns indicate rolling does not improve underlying signal quality

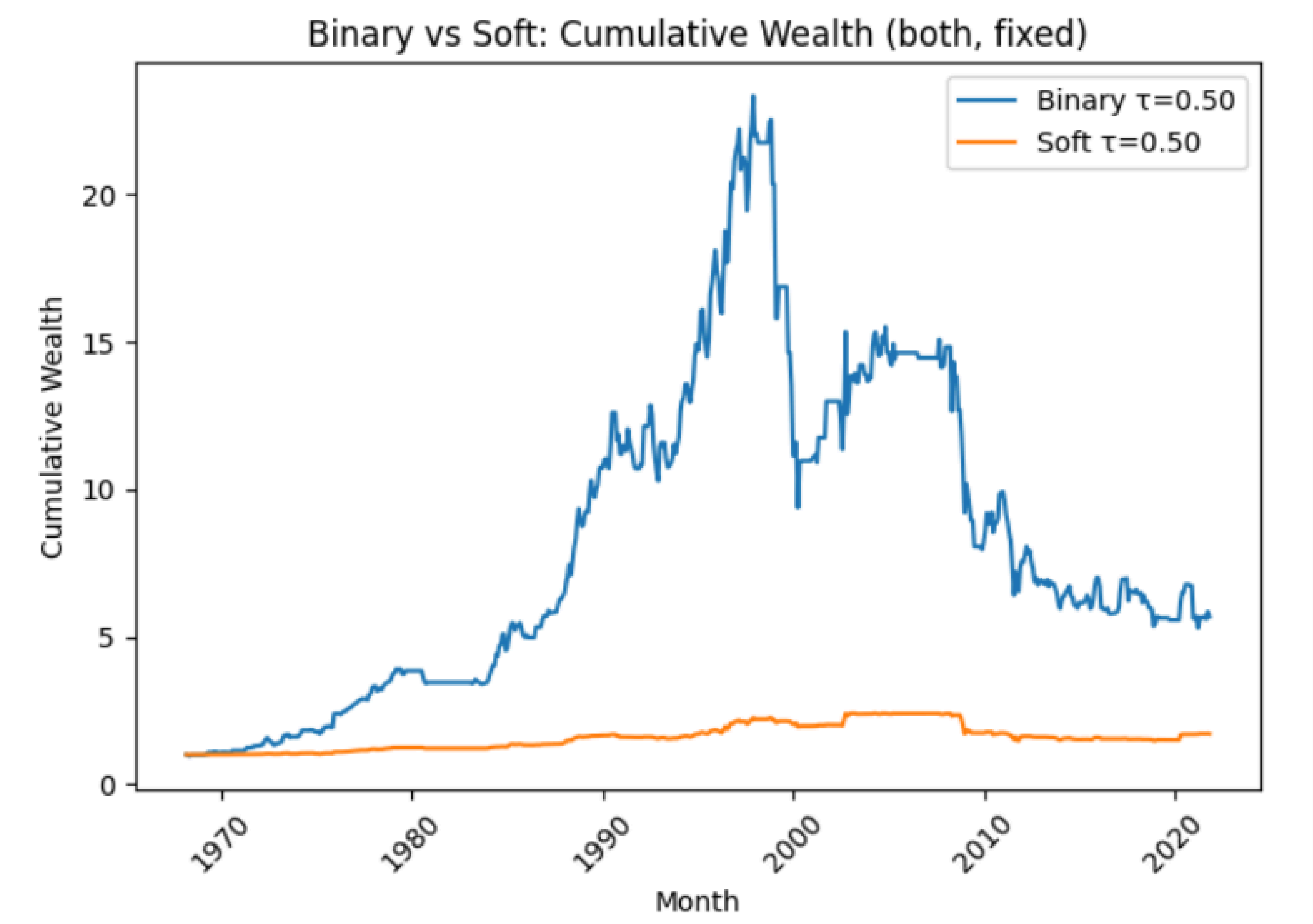
Annualized Volatility — Comparable volatility suggests risk levels are unchanged across training schemes

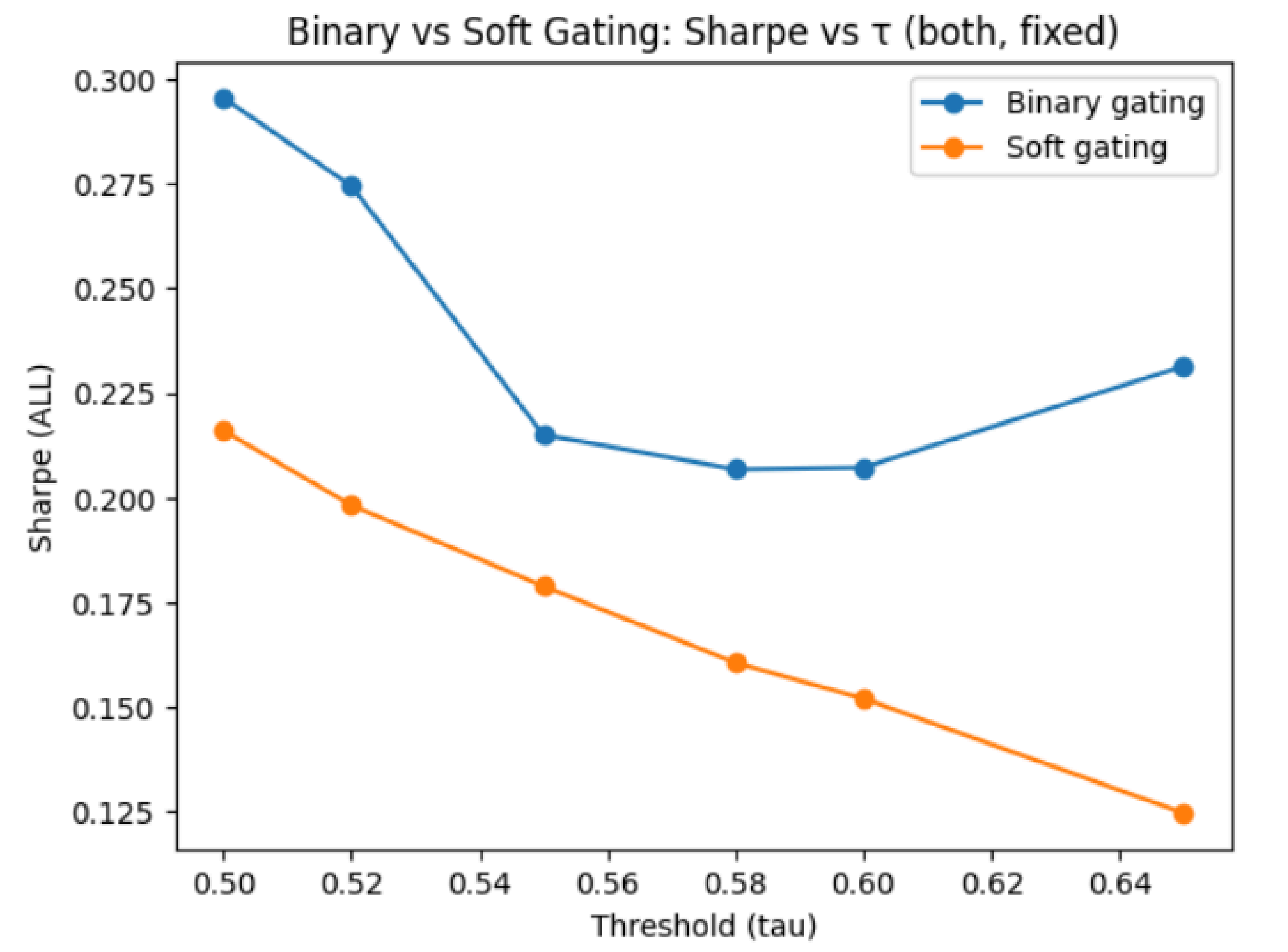
Maximum Drawdown — Fixed provides more consistent drawdown control, while rolling is less reliable

Trade Rate — Rolling produces more reactive and noisier exposure decisions over time

DESIGN ELEMENT 2 FINDINGS

Binary vs Soft Gating





Full Commitment Outperforms Scaled Exposure in Weak-Signal Settings

- Binary gating achieves higher wealth by allocating fully to selected trades.
- Soft gating underperforms due to under-allocation, dilutes returns without reducing risk.

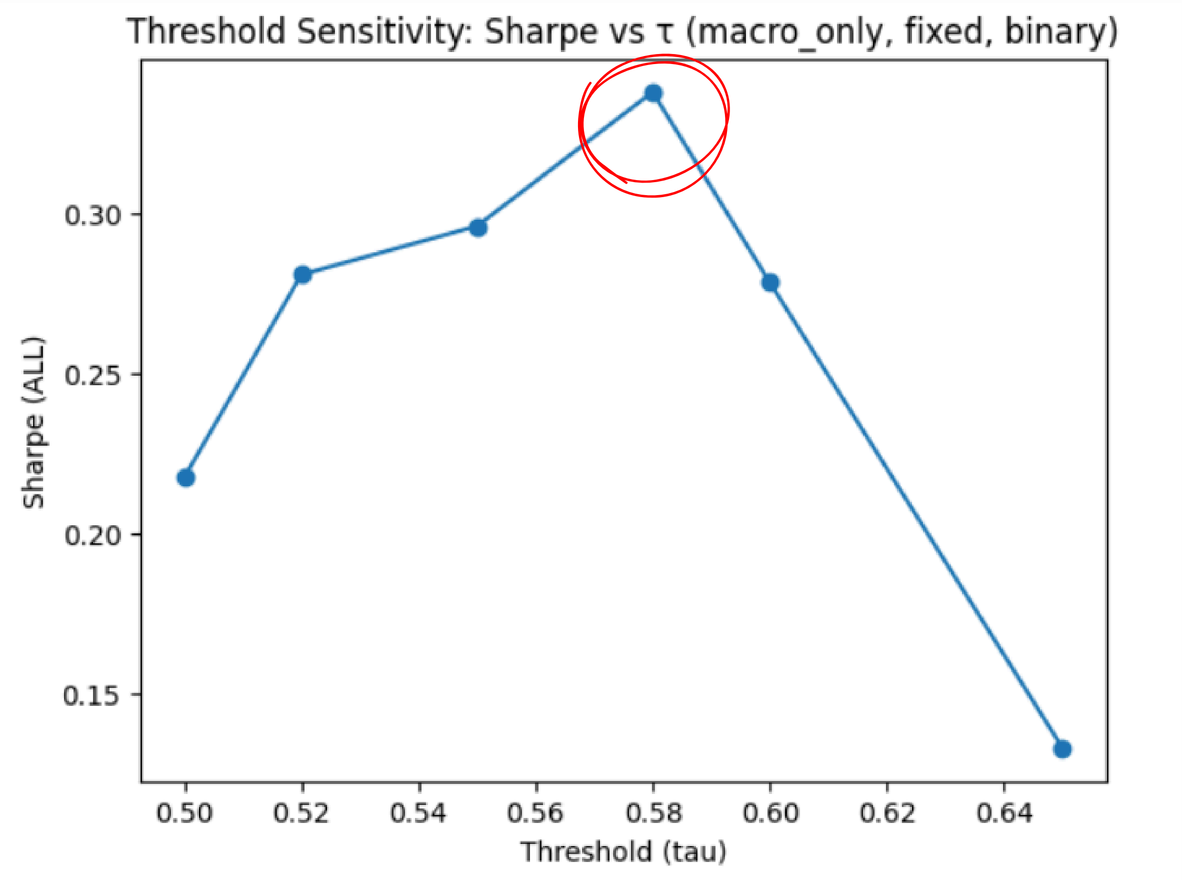
All-in decisions preserve signals better than scaling bets

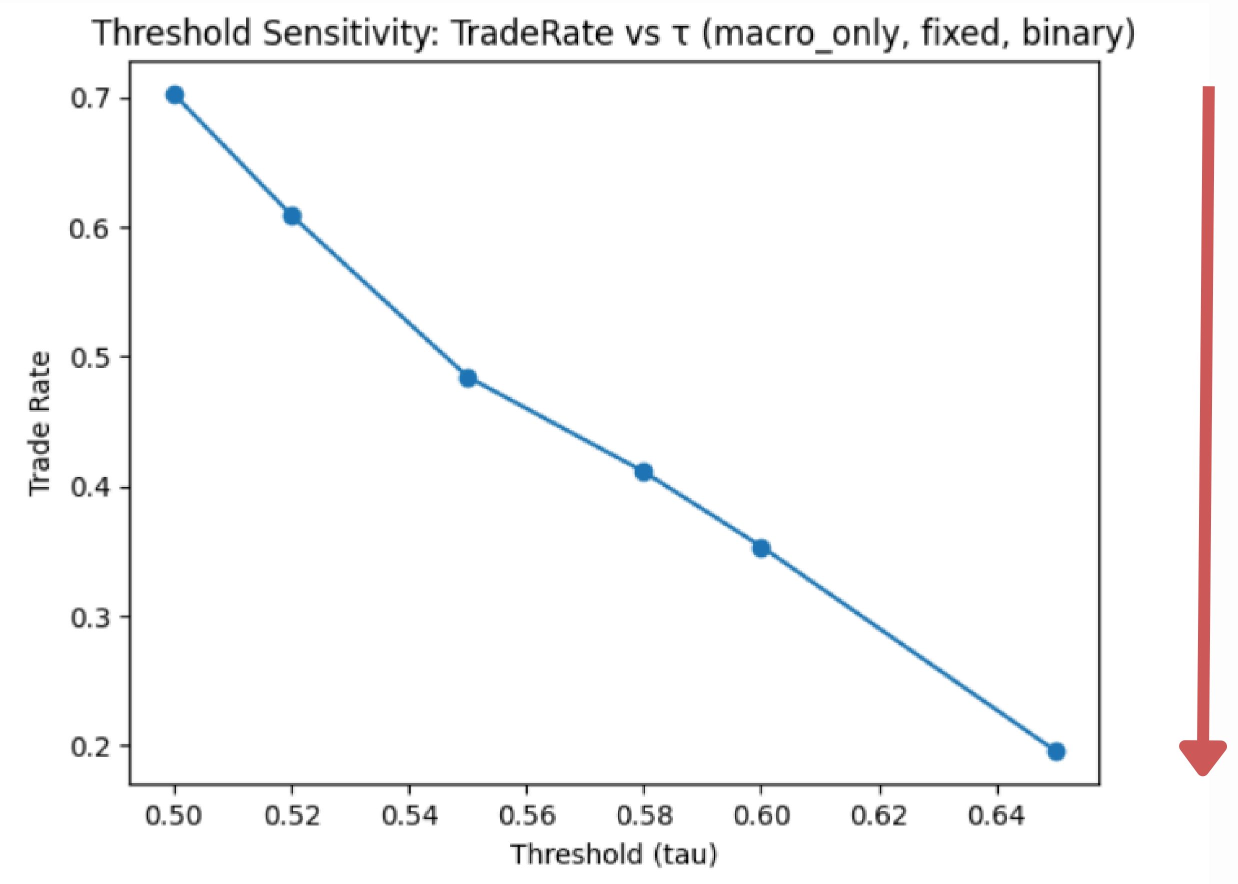
- Binary gating preserves signal strength by committing fully once signals pass the threshold, leading to consistently higher Sharpe.
- Soft gating becomes increasingly inefficient at higher τ, as fewer trades are taken and exposure is further diluted, compounding under-allocation.

DESIGN ELEMENT 3 FINDINGS

Varying Thresholds

Keeping the macro_only, binary, fixed model constant | τ ∈ {0.50, 0.52, 0.55, 0.58, 0.60, 0.65}





Non-linear performance relationship
Sharpe improves as τ increases due to filtering of low-confidence (noisy) signals. Beyond τ ≈ 0.58, performance declines as valid signals are discarded and diversification falls

Under-trading beyond optimal τ
At higher thresholds (τ ≥ 0.60), Sharpe deteriorates because the model filters out too many valid signals, leading to under-investment.

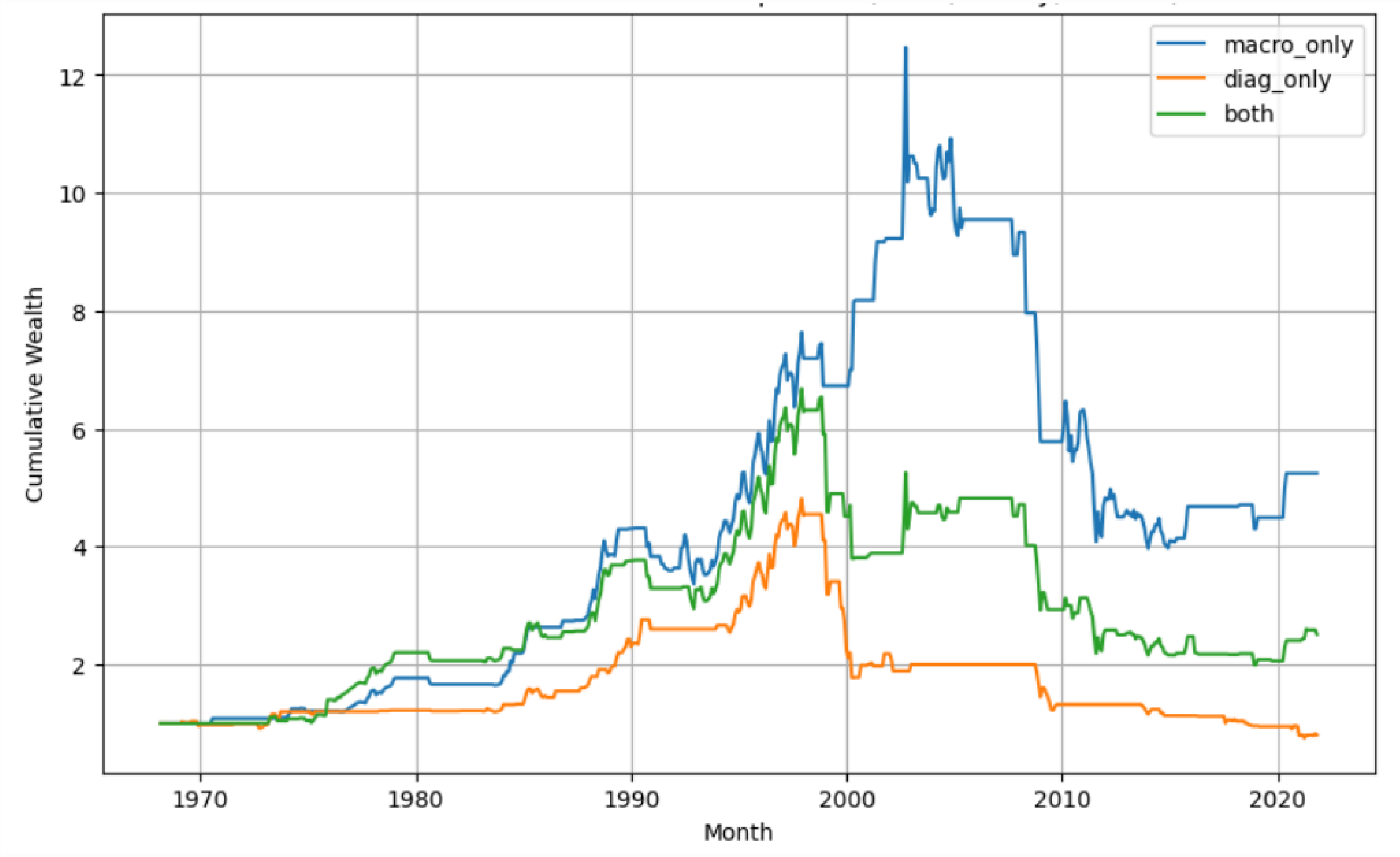
There is a Filtering vs opportunity tradeoff
As τ increases, trade rate drops significantly (≈70% → ≈20%), meaning the model becomes more selective but sacrifices diversification and opportunity.

DESIGN ELEMENT 4 FINDINGS

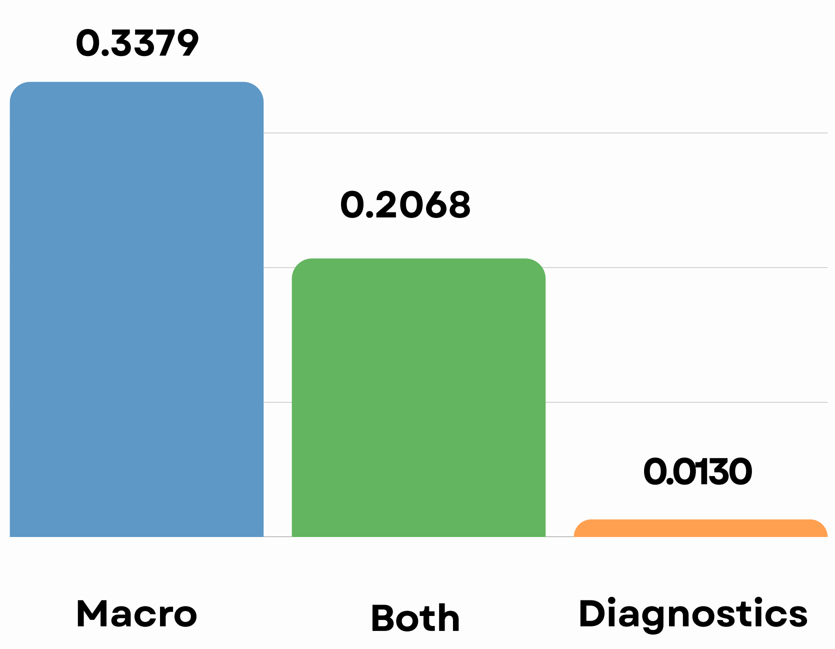
Feature Choice | Macro vs Diagnostics

Keeping the τ = 0.58, binary, fixed model constant

Cumulative Returns Comparison



Sharpe Ratio across feature inclusions



Macro signals dominate gating performance

- They provide the most reliable information for when to trade.

Adding diagnostics degrades performance

- More features introduce noise rather than improving decisions.

Diagnostics alone are insufficient

- Model-based signals are too unstable to drive gating decisions independently.

DESIGN ELEMENT 4 FINDINGS

Why Macro Outperforms Diagnostics

1. Macro captures true market regimes

Gating is fundamentally a “when to trade” problem, which is regime-driven, not stock-specific.

These signals are: stable over time, less noisy, directly aligned with portfolio-level risk

So the gate learns when the environment is favorable, which is exactly what it needs.

2. Diagnostics introduce noise rather than signal

The gating model tries to use both stable (macro) + noisy (diagnostics) signals. The noise distorts the decision boundary

This leads to worse filtering decisions & lower Sharpe

More information ≠ better decisions when the information is low quality

3. Mismatch between feature and objective

Diagnostics reflect model uncertainty, not market opportunity

So the gate ends up reacting to noise in predictions, it loses any connection to actual market regimes

Result: near-zero Sharpe

Chosen Gated Model: Fixed Macro Binary (τ = 0.58)

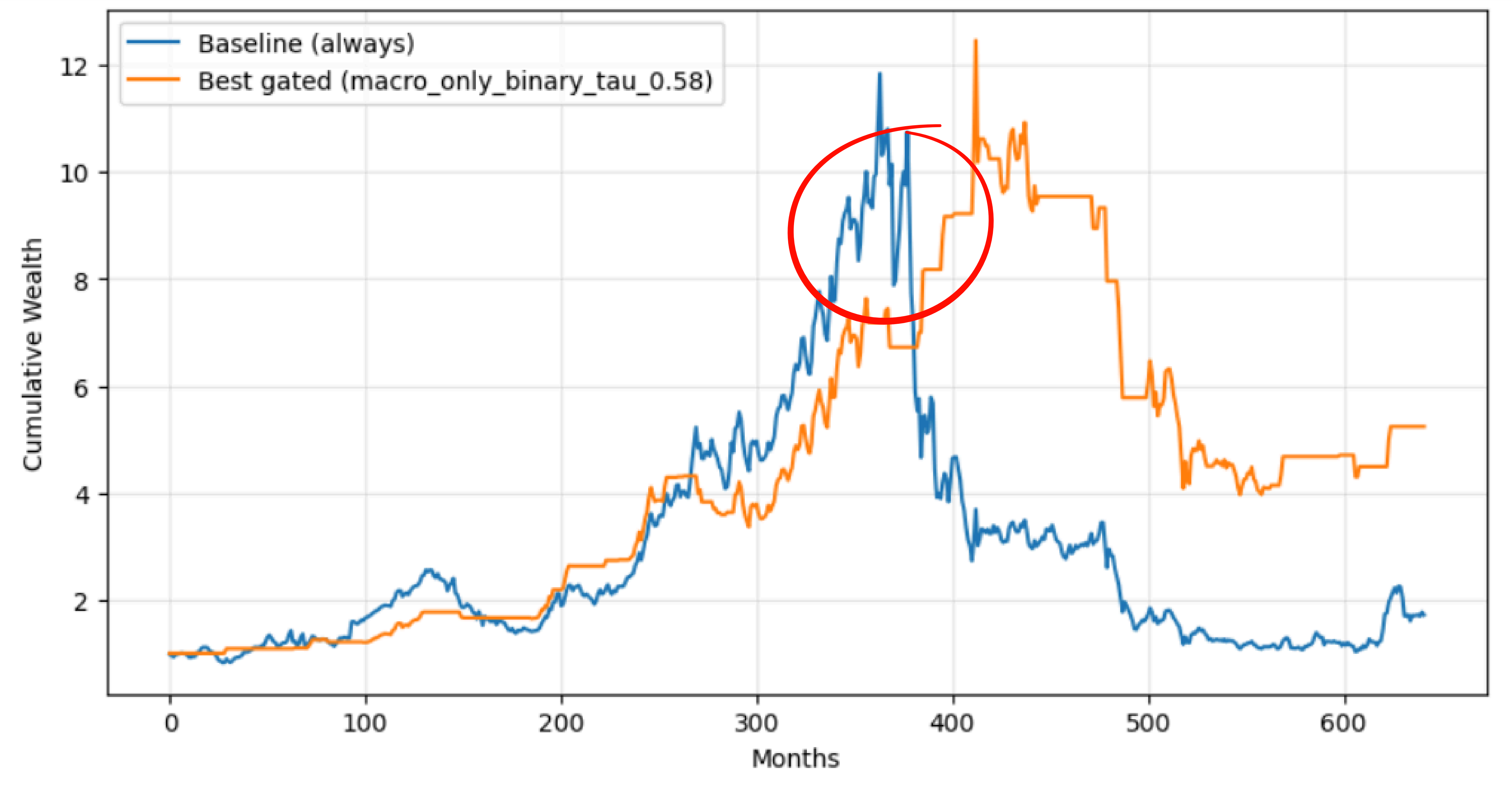
Overall, logistic regression was chosen due to its stability, interpretability & well-calibrated probabilities

Design Element	Theory Suggests	Our Choice (Why)
Training Scheme	Rolling adapts to changing markets	**Fixed** is more stable; **rolling amplifies noise** and reduces consistency
Feature Set	More features (macro & diagnostics) improve performance	**Macro-only** performs best; diagnostics introduce noise and degrade decisions
Gating Type	Soft gating optimizes risk via scaling	**Binary** preserves signal strength; soft under-allocates and weakens returns
Threshold (τ)	Higher τ improves signal quality	τ = 0.58 balances filtering noise and retaining opportunities

Decision Rule Estimate $p_t = P(R_{t+1} > 0)$. Trade only if $p_t \geq 0.58$, otherwise remain out

Crux | Does Gating actually improve performance?

Cumulative Returns Comparison

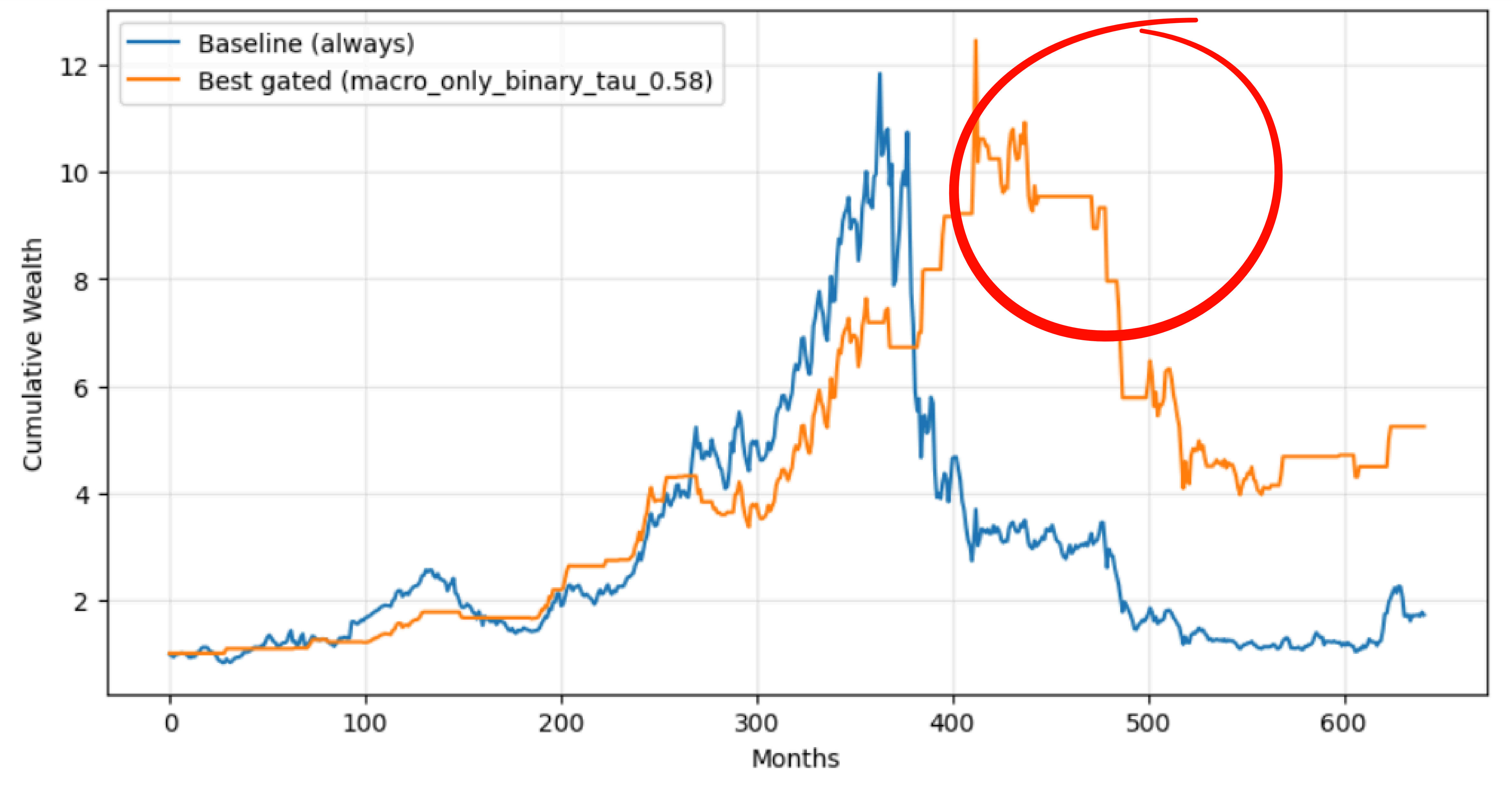


The meta-model struggles most during sharp regime transitions (e.g., ~350–400 months), where both strategies peak but the gated model still experiences drawdown.

This reflects timing uncertainty: probabilities are estimated using lagged macro signals, so when regimes shift abruptly, the model cannot adjust instantly and remains partially exposed.

Crux | Does Gating actually improve performance?

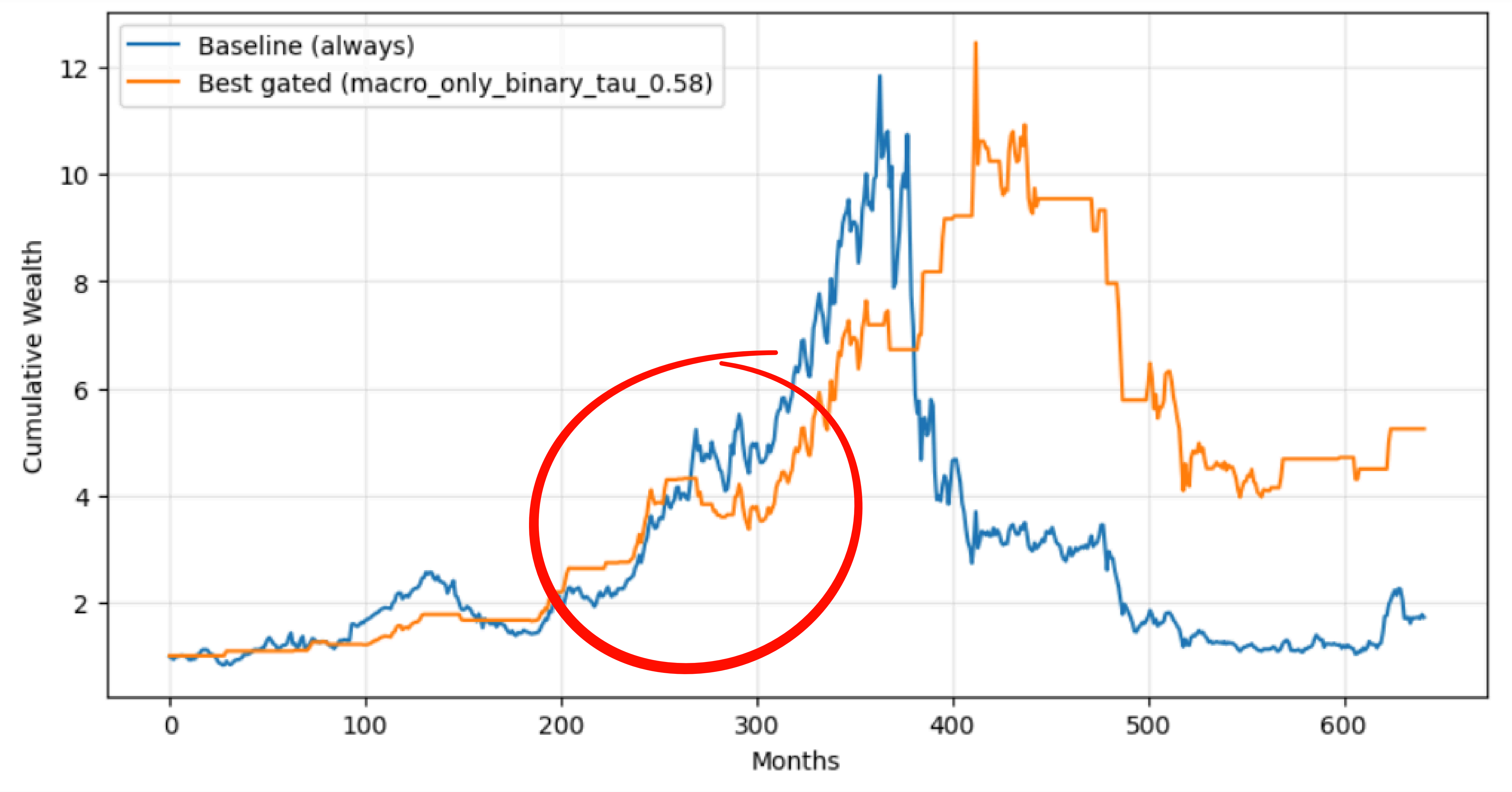
Cumulative Returns Comparison



The model performs strongest during prolonged deteriorations (post-400), where the gated strategy clearly avoids sustained losses
In these periods, macro signals become persistent and easier to classify, allowing the model to consistently assign low probabilities and reduce exposure effectively.

Crux | Does Gating actually improve performance?

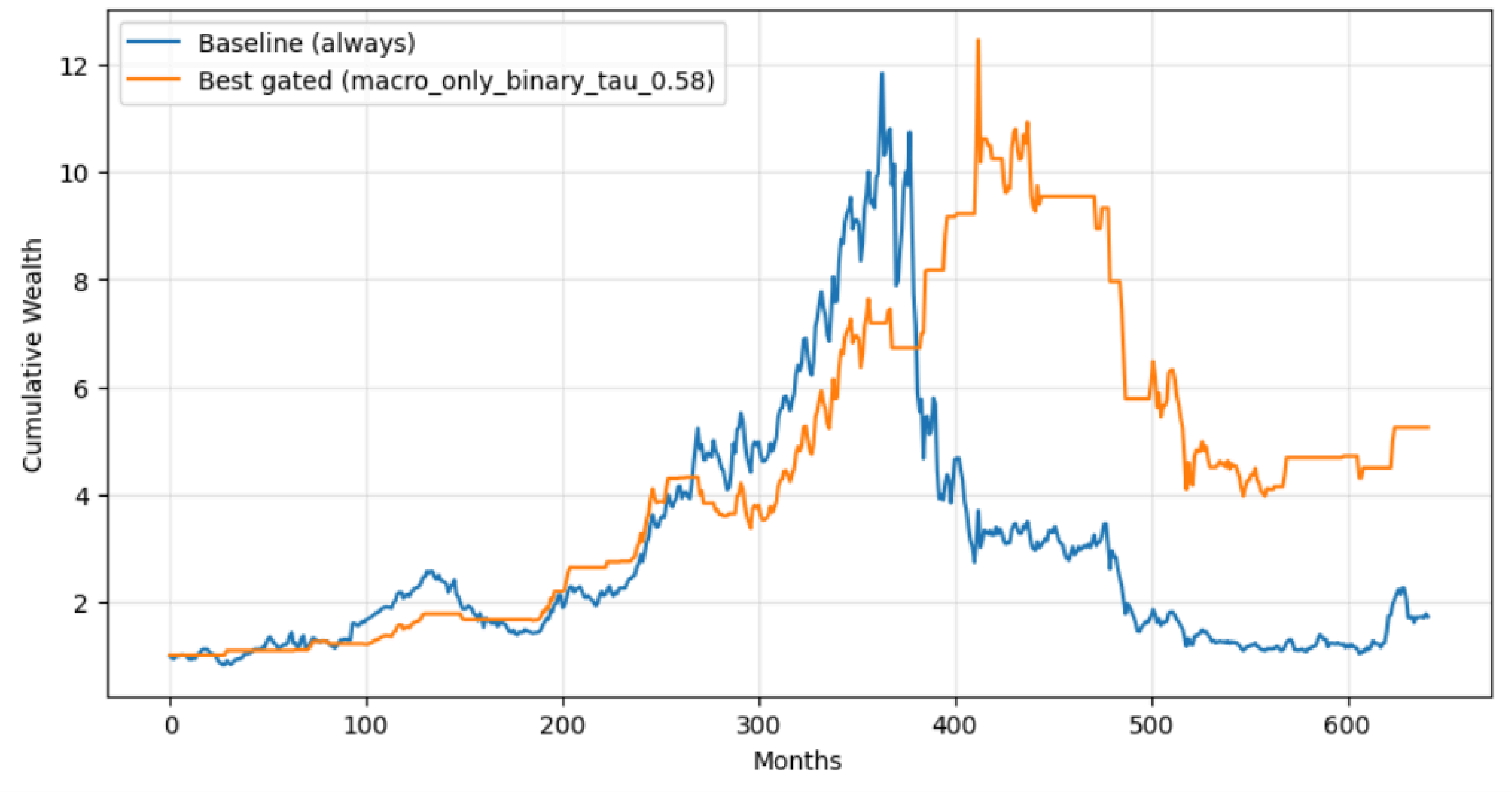
Cumulative Returns Comparison



The model sacrifices upside during strong bull runs (pre-350), indicating imperfect signal strength even in favorable regimes
Predicted probabilities do not consistently exceed the threshold (τ = 0.58), meaning the model is conservative under uncertainty, leading to under-participation despite positive returns.

Crux | Does Gating actually improve performance?

Cumulative Returns Comparison



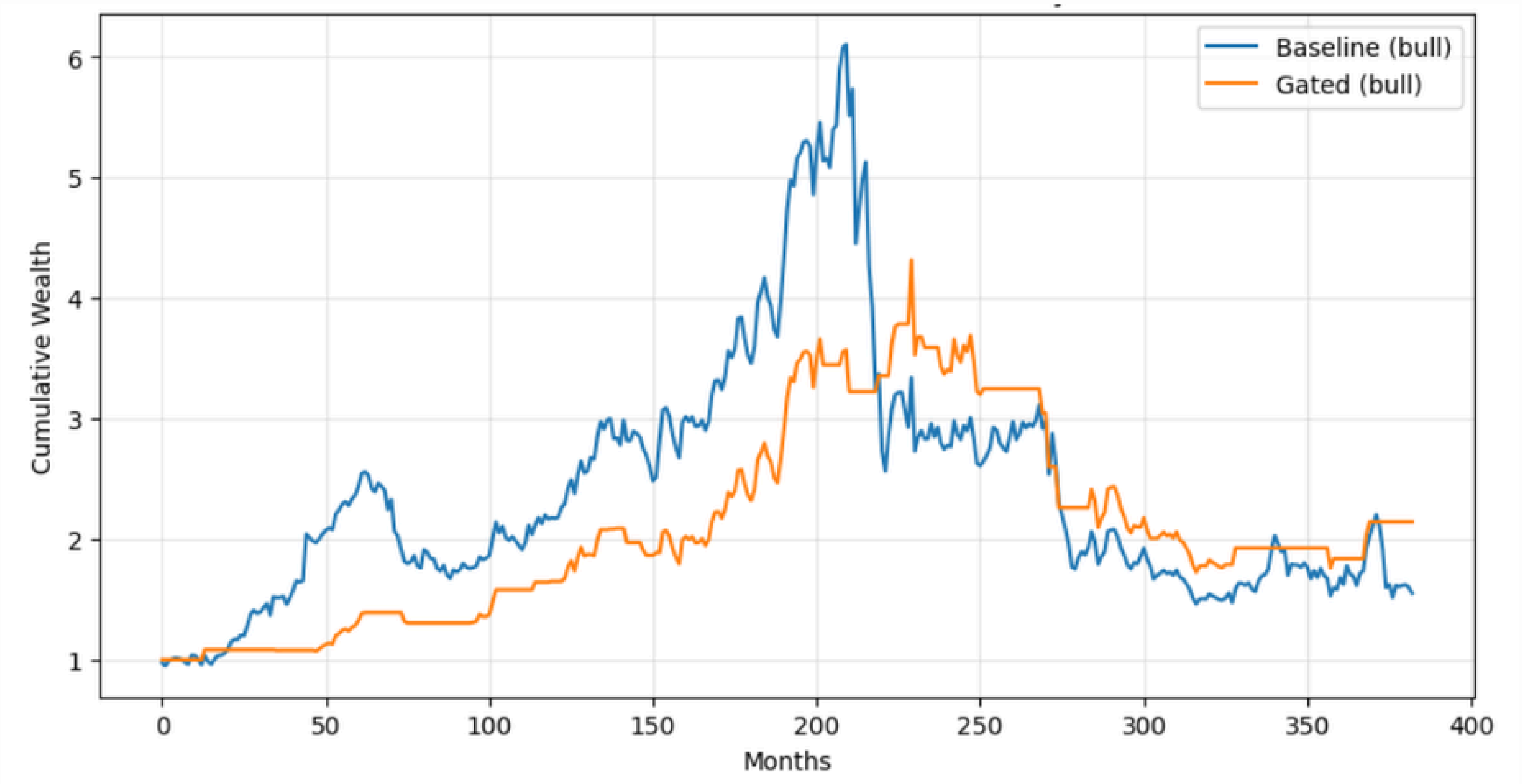
Despite noisy and imperfect regime prediction, gating still improves outcomes by asymmetrically avoiding large losses

While market regimes are hard to predict precisely, even partially informative signals are sufficient to improve performance because avoiding large drawdowns has a greater impact on long-term wealth than capturing every gain.

Crux | How is Gating actually changing behavior?

*Performance is regime-dependent, though Gating performs **slightly** better overall*

Bull Months | Cumulative Returns

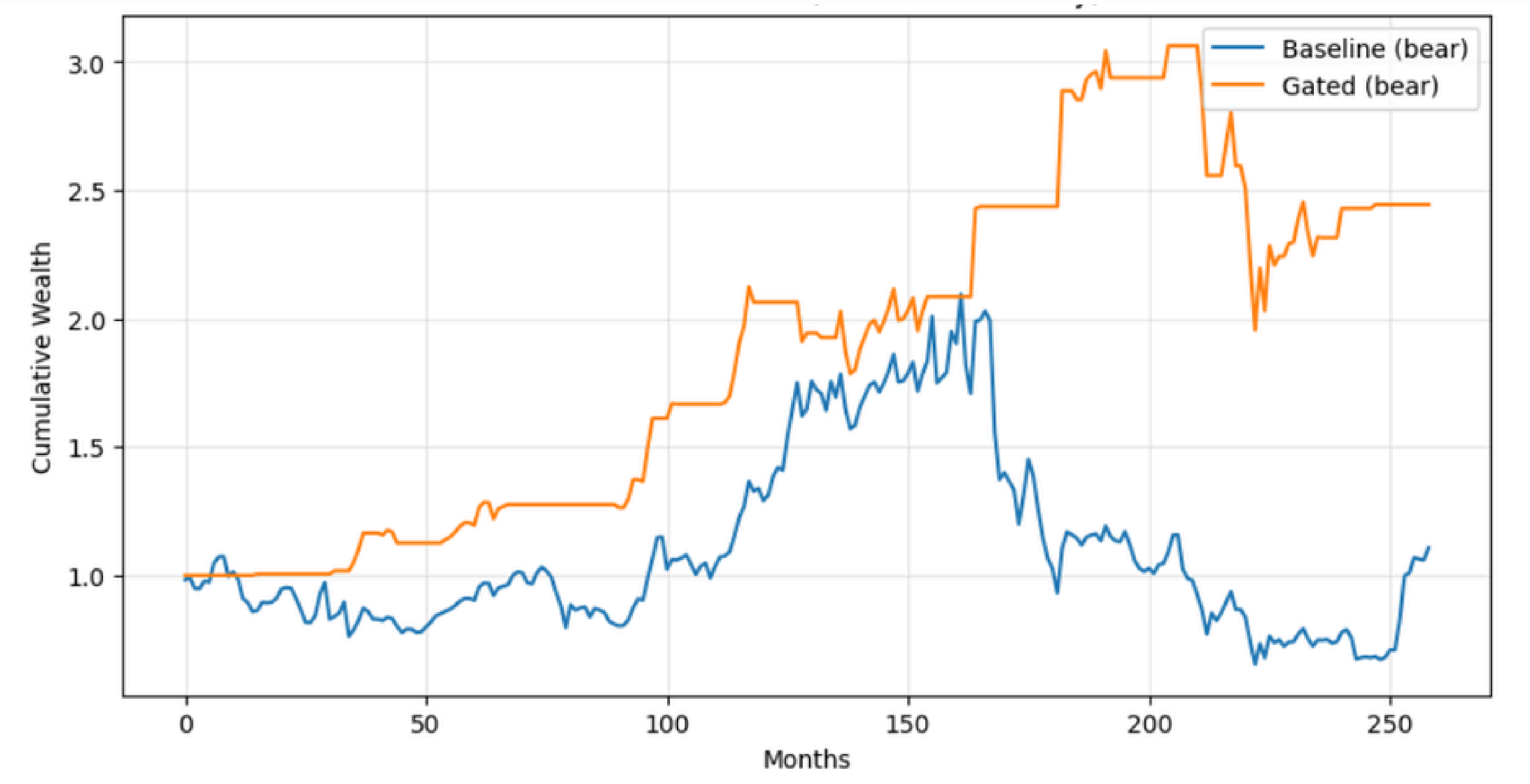


Gating underperforms the baseline during bull periods

The meta-model is conservative (τ = 0.58), so even in favorable regimes it fails to assign high enough probabilities consistently, leading to missed upside.

Occasionally outperforms by filtering out weak/noisy “bull” months

Bear Months | Cumulative Returns



Gating significantly outperforms during bear periods

Macro signals more reliably capture deteriorating conditions than positive ones, allowing the model to reduce exposure and avoid compounding losses.

Gating | Limitations & Practical Constraints

While the simple architecture improves stability, the primary limitation arises from weak and noisy signals rather than insufficient model complexity.

Weak regime separation
Bull vs bear probabilities overlap → gating decisions remain noisy and imprecise

Missed upside due to conservative threshold
$\tau = 0.58$ leads to under-exposure in strong bull periods → limits total return.

High sensitivity to threshold choice
Performance peaks narrowly → small changes in τ materially affect Sharpe and trade behavior.

Lag in adapting to regime shifts
Macro signals update slowly → model struggles at turning points and remains partially exposed.

Simple Model Architecture
Linear meta-model + binary gating cannot capture nonlinear dynamics or adjust exposure gradually.

Conservative training setup (5 epochs)
Model may be under-trained → potential underfitting and incomplete learning of available signal.

MGMT 59000 - Machine Learning
02/18/2026

Addendum A

Project Proposal

Task Definition & Data Choice

Empirical Asset Pricing
Predict next-month cross-sectional excess stock returns using a high-dimensional financial dataset

Research Task
"How does regime-aware exposure control affect the reliability assessment of trading signals?"

2-Stage ML Framework
Stage 1: Learn cross-sectional return signals with a regime-aware neural architecture.
Stage 2: Design & evaluate a lightweight regime-gating mechanism for dynamic exposure control

Evaluation Objective
Assess performance through a realistic long–short backtest and improvements in risk-adjusted returns.

Data Choice
Full historical dataset (1957-2021) since diverse economic environments & long training history are required

Validation Scheme

Time-Series Rolling Validation

Objective | Avoid look-ahead bias by using a rolling walk-forward validation scheme

Intuition: Simulating real world deployment

- Train the model using **10 years of past data**
- Use the **next 1 year** to validate and tune the model
- Slide the window forward by one month and repeat the train–validate–test cycle
- **Retrain model every year** using only information available at that time
- Evaluate via out-of-sample long–short portfolio performance (Sharpe, volatility, drawdown)

Baseline Model

Custom neural network with a multi-tower architecture

Basic Model Design

```
Macro → Tower \
Firm  → Tower  → Fusion → Return Prediction
SIC   → Tower /
```

Separate subnetworks for

- Macroeconomic variables (state of the economy)
- Firm characteristics (company fundamentals)
- Industry indicators (sector effects)

Trained using market-cap weighted loss function

$$\mathcal{L} = \sum w_{i,t}(\hat{r}_{i,t+1} - r_{i,t+1})^2$$

where $w_{i,t} \propto \mathrm{MarketCap}_{i,t}$

Evaluated Predictions by forming a monthly long-short portfolio

- Buy top predicted decile, short bottom predicted decile
- Evaluate out-of-sample performance via annualized return, volatility & Sharpe ratio

Meta-Model

Logistic Regression Classifier with a lightweight regime-gating layer

Basic Model Design

```
Macro State + Signal Diagnostics
          ↓
     Logistic Classifier
          ↓
      Exposure Gate
```

Inputs: macro state + signal diagnostics

Trained via binary cross-entropy (time-ordered split)

Gating Rule: Trade only if probability ≥ threshold, otherwise reduce / remove exposure

Research Focus

Whether signals are reliable across regimes

For this, we would change:

- Fixed vs Rolling Training
- Threshold sensitivity
- Exposure function (binary vs soft gating)
- Regime detection inputs

What we Learn

- How strict exposure control affects risk vs return
- Whether soft or hard gating better stabilizes performance
- How signal reliability varies across market regimes

Addendum B

Roadmap & Additional Insights

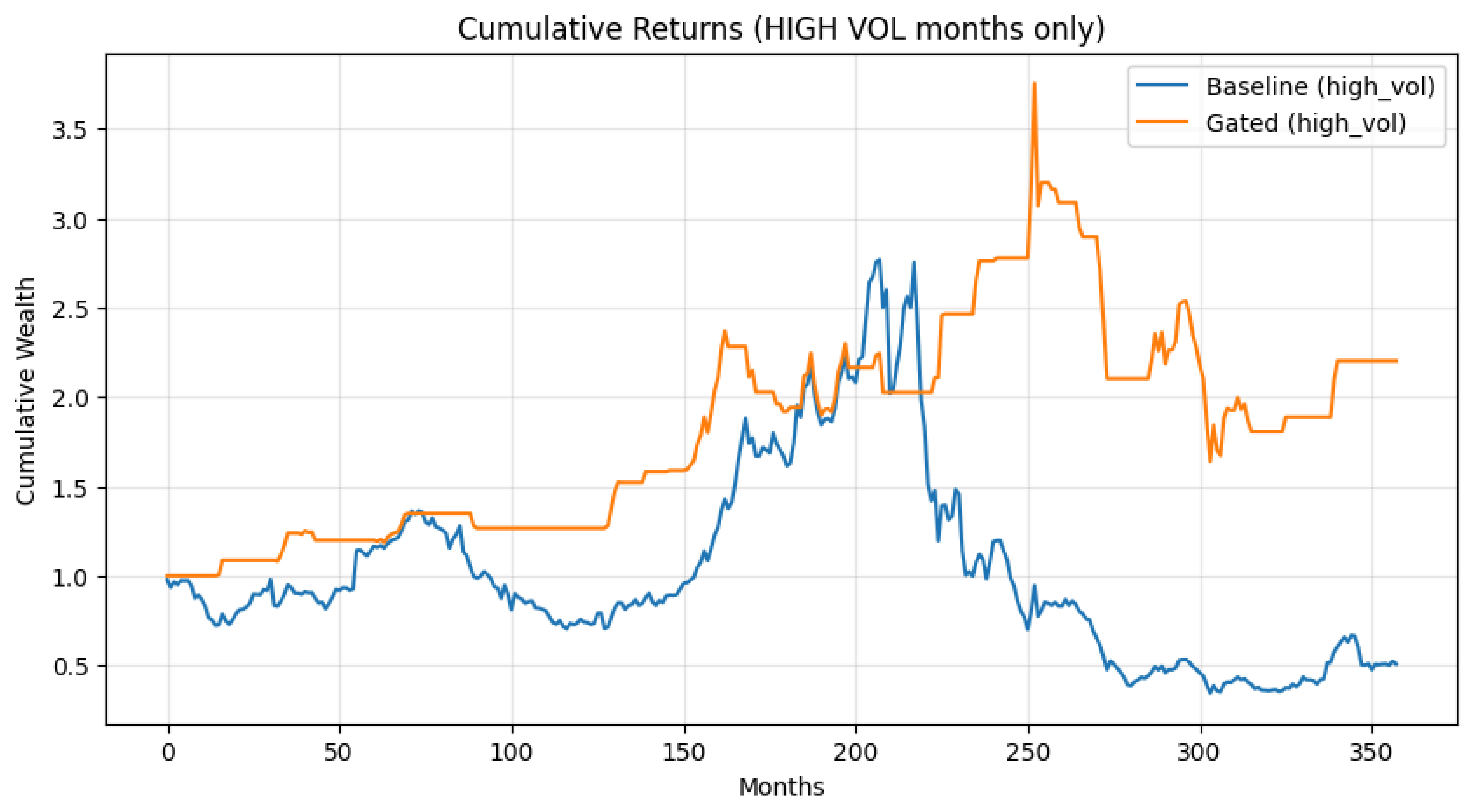
Roadmap

Baseline Return Prediction Strategy → Introducing Gating Layer → Gating Design Effects → Gating Presence Evaluation

1. We begin by establishing a baseline return prediction strategy.
2. We introduce a gating layer to control when signals are acted upon.
3. We analyze key design choices (training scheme, features, gating type, threshold) in terms of performance.
4. We evaluate whether gating improves performance and under what conditions it works

Our goal is not just to test whether gating works, but to understand when and why it works—or fails.

Impact of Gating on High Volatility Months



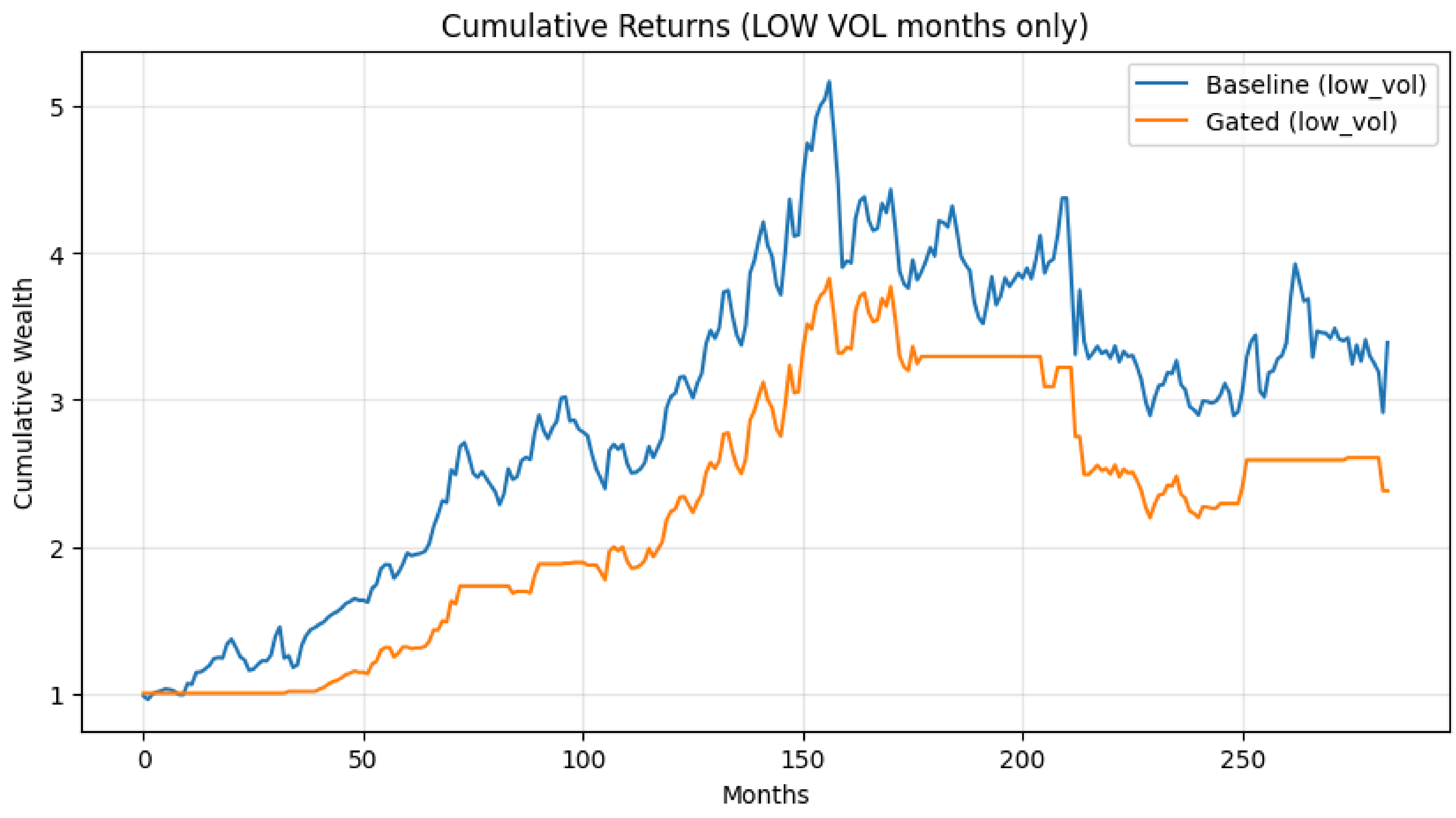
Interpretation

In high-volatility months, the gated model significantly outperforms the baseline, maintaining positive cumulative growth while the baseline experiences sharp declines and prolonged weakness. The divergence is most pronounced during crisis-like volatility spikes.

Why This Is Happening

High volatility often coincides with deteriorating macro conditions, which the gating model filters out; the baseline remains fully exposed and absorbs the volatility-driven losses.

Impact of Gating on Low Volatility Months



Interpretation

In low-volatility months, the baseline meaningfully outperforms the gated model, compounding more strongly and maintaining higher cumulative wealth. The gated strategy participates less consistently and therefore captures less of the steady gains typical of calm environments.

Why This Is Happening

Low volatility often reflects stable, favorable conditions where the strategy performs reliably; the always-invested baseline captures this fully, while the gated model's stricter confidence threshold causes it to forgo some profitable months.

MGMT 59000 - Machine Learning
02/18/2026

Thank you

Authors: Mishaal Hussain & Syed Bilal Farrukh